Teaching notes and solutions

> **Teacher note · Teacher notes.** S44 is placed before the first mock exam, but the logarithm is not part of its announced scope. Stress the conditions $a > 0, b > 0$ before combining logarithms. The differentiability of ln is accepted; the differentiation formula is proved. For a threshold with $0 < q < 1$, the negative sign of $\ln q$ reverses the inequality. Have students check the candidate value and the previous one.

Solutions to the activity

1. Each step multiplies the error by 0.7: $e_n = 5(0.7)^n$ for $n \geq 0$.
2. $e_n \leq 10^{-4}$ is equivalent to $(0.7)^n \leq 2 \times 10^{-5}$. Since ln is increasing, $n \ln 0.7 \leq \ln(2 \times 10^{-5})$. As $\ln 0.7 < 0$, we obtain
$$n \geq \frac{\ln(2 \times 10^{-5})}{\ln 0.7} \simeq 30.33512.$$
The first suitable integer is 31.
3. $e_{30} \simeq 1.1270 \times 10^{-4} > 10^{-4}$ and $e_{31} \simeq 7.8888 \times 10^{-5} < 10^{-4}$. The sequence of errors is decreasing, so this check confirms minimality.
4. For the target 10^{-6}, the same calculation gives $n \geq \frac{\ln(2\times10^{-7})}{\ln 0.7} \simeq 43.2465$, so $n = 44$. The real threshold increases by $\frac{\ln 100}{|\ln 0.7|} \simeq 12.91$, then we round up to the next integer. The number of steps is not multiplied by 100.

E1. We need $2x - 1 > 0$ and $x + 3 > 0$, so $x > \frac{1}{2}$. On this domain, the injectivity of ln gives $2x - 1 = x + 3$, that is $x = 4$. This value belongs to the domain; the solution set is $\{4\}$.

E2. We need $x > 2$. The product rule gives $\ln(x(x-2)) = \ln 3$, so $x^2 - 2x - 3 = 0$, that is $(x-3)(x+1) = 0$. Only $x = 3$ satisfies $x > 2$: the other root -1 is excluded.

E3. Since $x^2 + 1 > 0$, the first function is differentiable on $\mathbb{R}$ and its derivative is $\frac{2x}{x^2+1}$. On $]0, +\infty[$, the product rule gives $(x \ln x)' = \ln x + x(\frac{1}{x}) = 1 + \ln x$.

E4. For the first limit, set $t = \sqrt{x} \to +\infty$: $\frac{\ln x}{\sqrt{x}} = \frac{2 \ln t}{t} \to 0$. For the second, set $t = -\ln x \to +\infty$: $x^2 \ln x = -te^{-2t} \to 0$. Both limits use comparison of growth rates.

E5. For $x > 0$, set $g(x) = x - 1 - \ln x$. We have $g'(x) = \frac{x-1}{x}$, negative before 1, positive after. Its minimum is $g(1) = 0$, so $g(x) \geq 0$, which is equivalent to $\ln x \leq x - 1$.

Solutions to Homework 6

1. At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: $f \to -\infty$. At $+\infty$, $f = \frac{1}{x} + \frac{\ln x}{x} \to 0$. The sign is that of $1 + \ln x$: negative on $]0, e^{-1}[$, zero at e^{-1}, positive afterwards. By the quotient rule,
$$f'(x) = \frac{1 - (1 + \ln x)}{x^2} = -\frac{\ln x}{x^2}.$$
Thus f strictly increases up to 1, then strictly decreases; its maximum is $f(1) = 1$.

2. Differentiating $-(\ln x)x^{-2}$, we obtain $f''(x) = \frac{2\ln x - 1}{x^3}$. The numerator changes sign, from negative to positive, at $x = \sqrt{e}$. So f is concave on $]0, \sqrt{e}]$, convex on $[\sqrt{e}, +\infty[$ and has the point of inflection $(\sqrt{e}, \frac{3}{2\sqrt{e}})$.

3. On $]0, 1]$, continuity, strict increase, limit $-\infty$ and final value 1 give exactly one preimage of $\frac{1}{2}$. On $[1, +\infty[$, strict decrease from 1 towards 0 gives exactly one second preimage. The first is in $]e^{-1}, \frac{1}{2}[$, because $f(e^{-1}) = 0$ and $f(\frac{1}{2}) = 2(1 - \ln 2) \simeq 0.6137 > \frac{1}{2}$. The second is in $]5, 6[$, because $f(5) \simeq 0.5219 > \frac{1}{2} > f(6) \simeq 0.4653$.

4. The maximum gives $\frac{1+\ln x}{x} \leq 1$. Since $x > 0$, we can multiply without changing the direction: $1 + \ln x \leq x$. Equality corresponds to the unique maximum of f, so to $x = 1$.

Solutions to the textbook or past-paper work

The logarithm requires a strictly positive argument.

1. $3x - 7 > 0 \iff x > \frac{7}{3}$, so $D_f = (\frac{7}{3}, +\infty)$.

2. $-x^2 + 4x - 3 = -(x-1)(x-3) > 0$ exactly when $1 < x < 3$. Thus $D_g = (1, 3)$.

3. We need simultaneously $x > 0$ and $2 - x > 0$; their intersection gives $D_h = (0, 2)$. The coefficient -3 placed in front of the second logarithm does not change its condition of definition.